

AnchorIQ

Customer Churn & Retention Analysis

Dataset: telco_churn.csv

Generated: October 03, 2026 - 03:14 | Powered by AnchorIQ v2.0

7,043 Records	21 Features	Churn Target	0 Fixed Nulls	0 Dupes Cleared
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1. EXECUTIVE SUMMARY & AI INSIGHTS

Executive Summary

We analyzed 7,043 customer records to predict churn. The overall churn rate is 26.5 %. A logistic regression model (AUC = 0.8414) identifies the 10 % of customers most likely to leave, capturing 29 % of all churn cases. Several customer attributes—short tenure, month-to-month contracts, fiber optic service, and higher charges—are strong churn drivers. Targeted retention on these groups can reduce churn significantly.

Key Findings

- Overall churn:** 1,869 of 7,043 customers (26.5 %).
- Tenure 0-9 months:** 49.8 % churn (1.9× higher than overall) across 1,854 customers (26 % of the base).
- Month-to-month contract:** 42.7 % churn (1.6× higher) among 3,875 customers (55 % of the base).
- Fiber optic Internet:** 41.9 % churn (1.6× higher) in 3,096 customers (44 % of the base).
- Top 10 % risk score:** 704 customers contain 29 % of all churn cases—2.9× better than random selection.

Recommendations

- Act on the 704 highest risk customers**

Why: They hold 29 % of churn cases.

Impact: Targeted offers to this group could capture a large share of churn with minimal outreach.

Assumption: A 10 % relative drop in churn for this group reduces ~54 churn cases.

- Target month-to-month contract holders**

Why: 42.7 % churn rate, 1.6× higher than average.

Impact: A 10 % relative reduction would save

2. DATASET OVERVIEW & QUALITY AUDIT

Audit Metric	Value	Status
Total Records	7,043	Validated
Total Feature Count	21	Validated
Numerical Columns	3	Active
Categorical Columns	18	Active
Target Column	Churn	Identified
Missing Cells Imputed	0	Resolved
Duplicate Rows Purged	0	Resolved
Data Quality Index	100.0%	Optimal

Preprocessing Pipeline Execution:

- Converted 'TotalCharges' to numeric
- Filled 11 nulls in 'TotalCharges' with median (1397.47)

3. MACHINE LEARNING BENCHMARKS

Problem Formulation: **Classification** | Best Champion: **Logistic Regression** | Training: **5,634** samples | Evaluation: **1,409** samples | Engineered Features: **19**

Model Candidate	AUC Score	Validation Status
Champion: Logistic Regression	AUC = 0.8414	Top Performer
Logistic Regression	AUC = 0.8449	Linear
Random Forest	AUC = 0.8423	Ensemble
XGBoost Classifier	AUC = 0.8408	Gradient Boosted
Accuracy	73.74%	Overall Match
Cross-Validation	0.8449 ± 0.006	5-Fold CV

Key Drivers & Influential Signals:

Rank	Feature Variable	Importance Weight
1	tenure	0.4783
2	InternetService	0.1578
3	Contract	0.1155
4	MonthlyCharges	0.1076
5	TotalCharges	0.0386
6	StreamingMovies	0.0181
7	StreamingTV	0.0171
8	OnlineSecurity	0.0140

4. EXPLORATORY DATA ANALYSIS (EDA CHARTS)

Figure 1: Feature Distributions

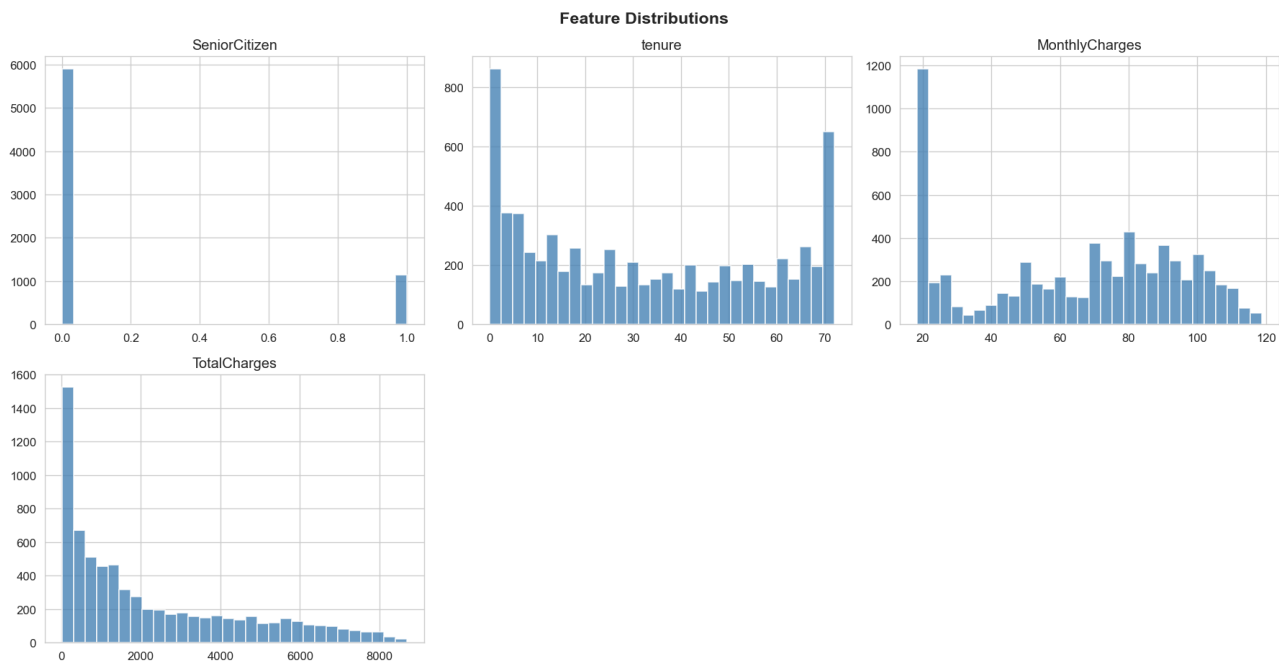


Figure 2: Feature Correlation Heatmap

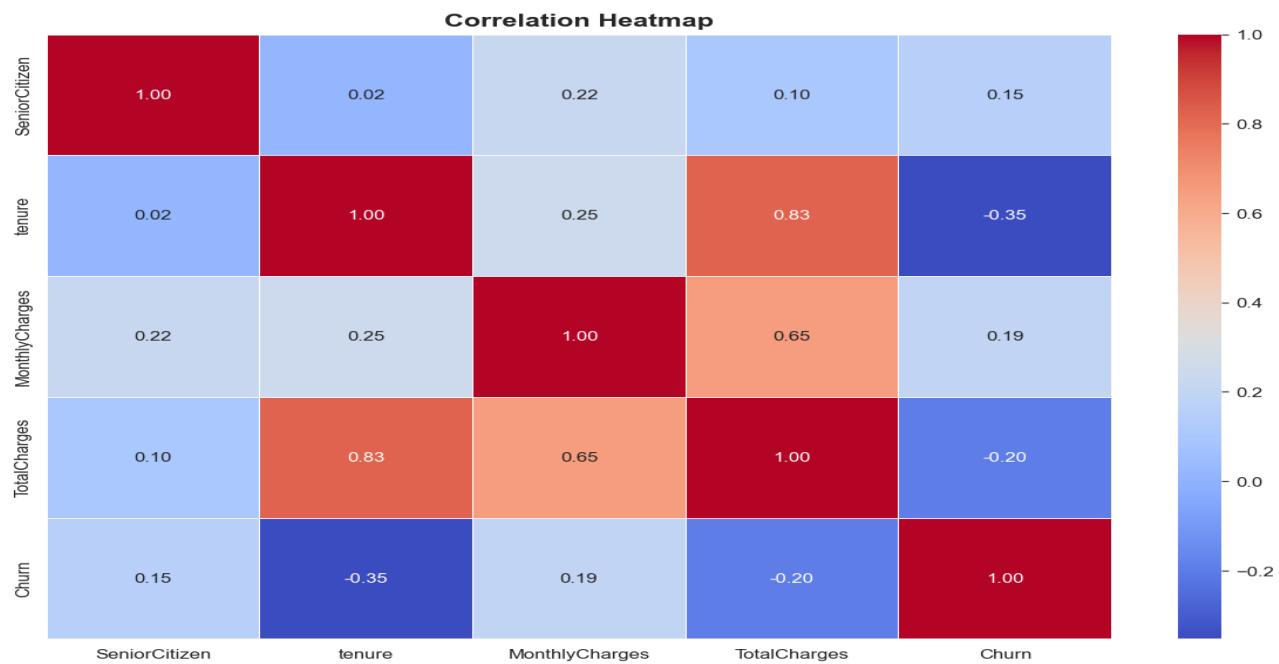


Figure 3: Target Variable Distribution

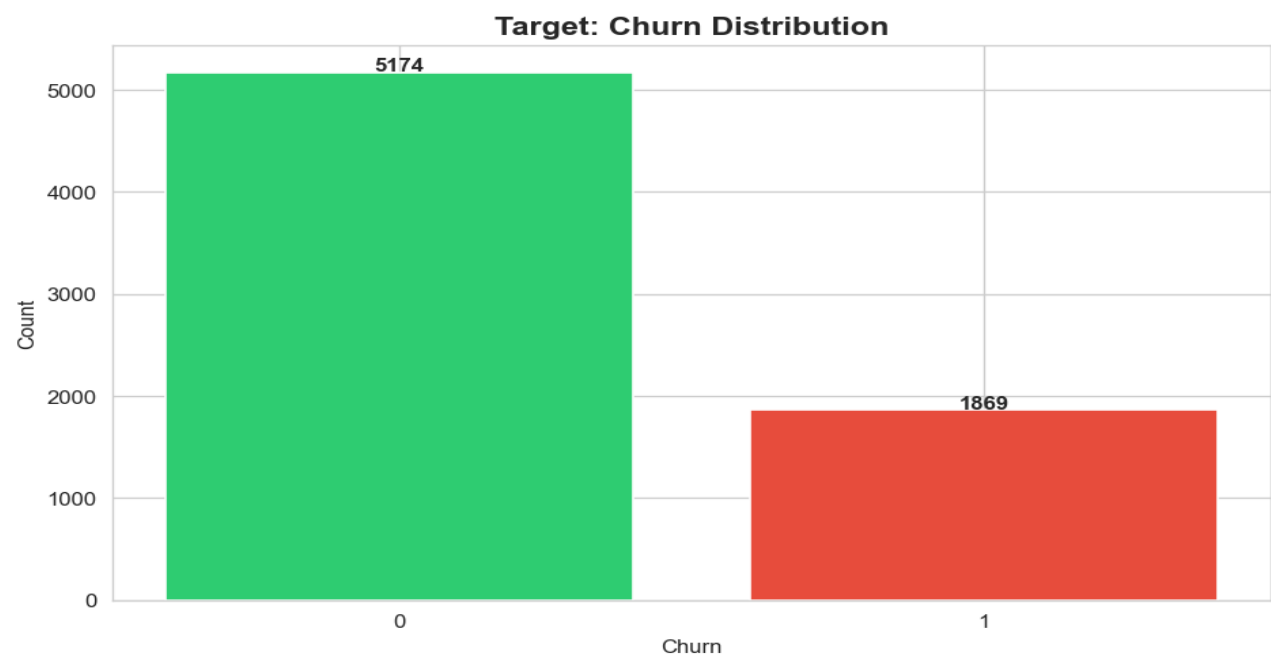


Figure 4: Categorical Feature Impact

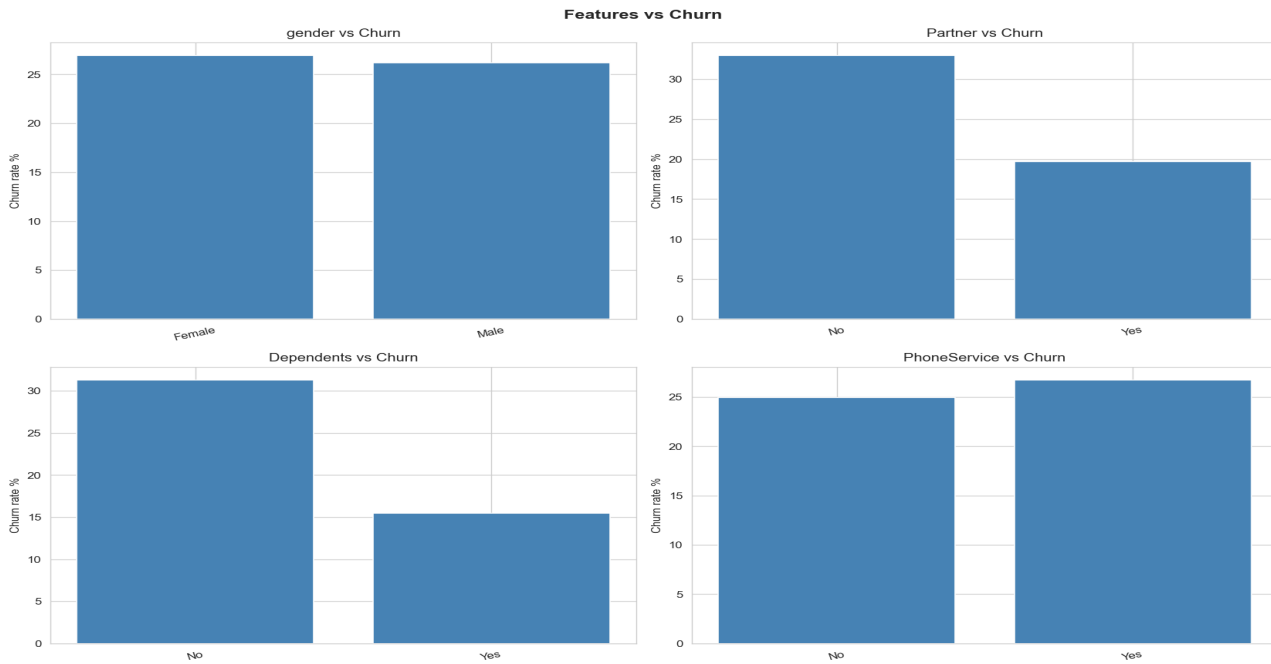
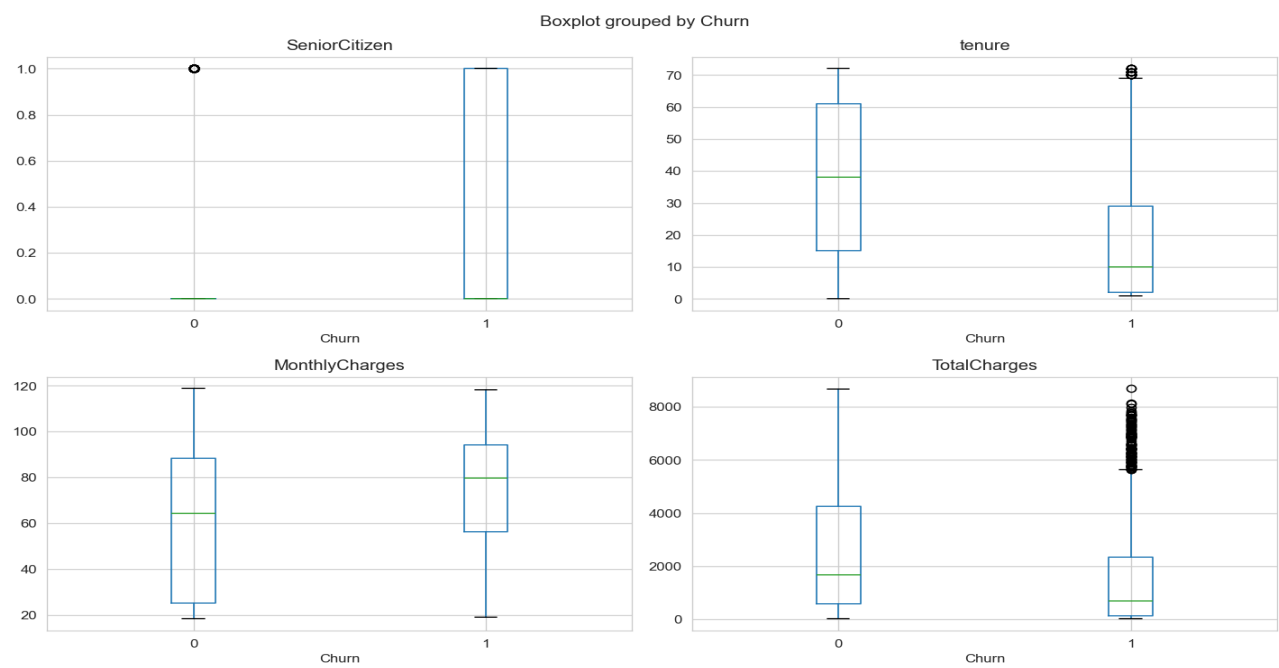


Figure 5: Numerical Distributions by Target



5. FEATURE STATISTICAL SUMMARY

Feature	Mean	Median	Std Dev	Min	Max	Skewness
SeniorCitizen	0.16	0.0	0.37	0.0	1.0	1.83
tenure	32.37	29.0	24.56	0.0	72.0	0.24
MonthlyCharges	64.76	70.35	30.09	18.25	118.75	-0.22